Rongfeng **Cheng** | Medical AI Research

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Honors & Awards

2025: Campus Gold Award, Internet+ Competition – Dingbei Xinxin cardiovascular health-management project.

2025: Second Prize, 9th Guangzhou Mathematical Modeling University League – fruit classification with ConvNeXt.

2025: Third Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Division – nonlinear programming, genetic algorithms, and particle-swarm optimization.

Education

South China University of Technology (SCUT) **Guangzhou, China**
Bachelor of Management in Big Data Management and Application *2023–2027*
GPA: 3.46/4.0.
Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Research & Project Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

DLEF: Dataset-Label Evidence Fingerprints
Jun 2026–Present
Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman $r = 0.843$ across 108 points; $r = 0.892$ after source-target-label aggregation). See Manuscript [1].

Evidence-Use Auditing for Deep ECG Models
Jan 2026–Present
Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions. See Manuscript [2].

Dingbei Xinxin / GradioGPT: Proactive Cardiovascular Health Management LLM
Co-founder & Core Developer; Director since Aug 2026 *2025–Present*

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

Myocardial Ischemia Knowledge Graph
Project Lead, SCUT Student Research Program *2025*
Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

IHD Trajectory Similarity Retrieval
Independent Research *Nov–Dec 2025*
Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays in MIMIC-IV using time-windowed clinical variables and precomputed k-NN similarity graphs.

Manuscripts & Publications

Under Review [1]: **R. Cheng et al.** "DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification." *IEEE BIBM 2026*. First author.

Under Review [2]: **R. Cheng et al.** "A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification." *IEEE JBHI*. First author.

Ongoing: **PathCLIP: Evidence-Aware ECG–Text Representation Learning.** Ongoing first-author study of ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

Apr 2026 [3]: J. He, S. Qin, H. Zou, F. Jing, W. Liu, C. Li, Q. Wang, Z. Hu, K. Song, Z. Xu, Q. Luo, Y. Cheng, **R. Cheng**, H. Zhang, and W. Cheng. "CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation." *iScience*. doi:10.1016/j.isci.2026.115073.

Mar 2026 [4]: R. Wang, F. Jing, **R. Cheng**, Y. Cheng, W. Cheng, H. Zou, H. Ren, and W. Jin. "Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques." HNNDL 2026. doi:10.1145/3795892.3795917.

Mar 2026 [5]: Y. Fang, F. Jing, **R. Cheng**, Y. Cheng, W. Cheng, H. Ren, W. Cai, and W. Jin. "Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification." HNNDL 2026. doi:10.1145/3795892.3795918.

Internship Experience

AI Research Institute, Guangdong Second Provincial General Hospital
LLM Development Engineer Intern *Nov 2024–Present*

- Developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

Guangzhou Yunzhan Zhichuang Information Technology Co., Ltd.
Backend & Algorithm Development Intern *Jul 2025–Jan 2026*

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company's iTwins product.

Patents & Software IP

Patent: **Method, System, and Electronic Device for Medical Question-Answering Output Assurance** – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

Software IP: Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

Skills

Modeling: PyTorch, XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

LLM & Knowledge Systems: LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

Evaluation & Data Curation: RAG/retrieval evaluation, perturbation analysis, external validation, schema design, data harmonization

Engineering: Python, TypeScript, SQL, Neo4j, Next.js, Flask, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528